

Linearly Independent Row Vectors and Rank

Lecturer: Chng Zhi Yee (zy_chng@um.edu.my)
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Learning Outcomes

At the end of this lecture, you should be able to:

- ☐ Know what is linearly dependent and linearly independent.
- ☐ Determine if a given vector set is linearly dependent or linearly independent.
- ☐ Determine the rank of a matrix.

Linearly dependent and independent

Let $v_1, v_2, \dots, v_n$ be n vectors in a vector set V ($V = \{v_1, v_2, \dots, v_n\}$).

(i) $v_1, v_2, \dots, v_n$ are **linearly independent** if

$$\lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n = 0 \text{ implies } \lambda_1 = \lambda_2 = \dots = \lambda_n = 0.$$

(ii) $v_1, v_2, \dots, v_n$ are **linearly dependent** if

there exist numbers $\lambda_1, \lambda_2, \dots, \lambda_n$, not all of them zero,
such that $\lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n = 0$.

Linearly dependent and independent

Example:

Show that the set $V = \{(1, 0, 0), (1, 2, 0), (1, 0, 1)\}$ is linearly independent.

Linearly dependent and independent

Example:

Show that $V = \{(1, 0, 0), (1, 2, 2), (1, -4, 4)\}$ is a linearly dependent set.

Linearly dependent and independent

Let $V = \{v_1, v_2, \dots, v_n\}$. If there exists a zero row vector in V , then $v_1, v_2, \dots, v_n$ are linearly dependent.

Rank

Let $A = [a_{ij}] \in M_{m,n}(\mathbb{R})$ and let v be a row vector of A .

Then, the maximum number of independent row vectors of A is called the **rank** of A and is denoted by $\rho(A)$ or **rank** A .

Example:

Find the rank of $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 2 \\ 1 & -4 & -4 \end{bmatrix}$.

Rank

Let $A \in M_{m,n}(\mathbb{R})$ and $A \in M_{n,q}(\mathbb{R})$. Then

- (i) $\rho(AB) \leq \min\{\rho(A), \rho(B)\}$.
- (ii) $\rho(PAQ) = \rho(A)$ for every invertible matrices $P \in M_m(\mathbb{R})$ and $Q \in M_n(\mathbb{R})$.

Rank

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Rank

If $A = [a_{ij}] \in M_{m,n}(\mathbb{R})$ is a row echelon matrix, then the rank of A , $\rho(A)$ = the number of non-zero row(s) in A .

Let A be a matrix formed by the row vectors set V .

(a) $V_1 = \{(0, 1, 0), (1, 0, 2), (2, 0, -1)\}$.

(b) $V_2 = \{(1, -1, 4, 2), (0, 1, 3, 2), (3, -2, 15, 8)\}$.

(i) Determine the rank of A .

(ii) Determine whether the row vectors set V is linearly dependent or is linearly independent.

(iii) Determine the maximum number of linearly independent row vectors in V .

Rank